



Roll No:

MPHARM**(SEM. I) THEORY EXAMINATION 2025-26****DRUG DELIVERY SYSTEM****TIME: 3 HRS****M.MARKS: 75****Note: 1.** Attempt all Sections. If you require any missing data, then choose suitably.**SECTION A****1. Attempt all questions in brief.****10 x 2 = 20**

a.	State the demerits of SR/CR formulations.
b.	Define pharmacogenetics and mention its 2 major applications.
c.	State the principles of feedback-regulated drug delivery systems.
d.	Mention the role of the GI transit time approach on gastroretentive drug delivery.
e.	Name 2 polymers used for mucoadhesion.
f.	Mention the significance of intracorneal drug delivery.
g.	Draw a schematic diagram representing skin barriers to drug delivery.
h.	State the mechanisms of drug permeation through skin.
i.	Mention the difficulties for protein drug delivery.
j.	Define and classify vaccines with examples.

SECTION B**2. Attempt any two parts of the following:****2 x 10 = 20**

a.	Explain the stages and processing involved in Telepharmacy.
b.	Discuss the principle, merits, demerits, mechanism of drug permeation, and applications of mucoadhesive drug delivery systems.
c.	Classify the ophthalmic barriers and explain their negative roles in ophthalmic drug delivery, with a neat schematic diagram.

SECTION C**3. Attempt any five parts of the following:****5 x 7 = 35**

a.	Explain the mechanisms of drug delivery through SR/CR formulations.
b.	Analyze the role of bioelectronic medicines in customization.
c.	Write down the principles, merits, demerits, and applications of pH-activated DDS.
d.	Classify gastroretentive drug delivery systems and discuss their formulation approaches.
e.	Elaborate on the role of various penetration enhancers with examples, for improved transdermal drug delivery.
f.	Propose the formulation approaches to overcome the difficulties faced during protein drug delivery.
g.	Justify the need and role of single-shot vaccines.